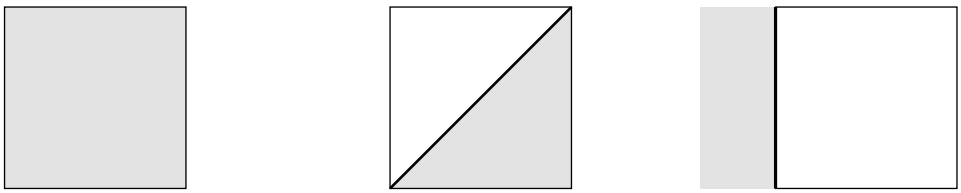
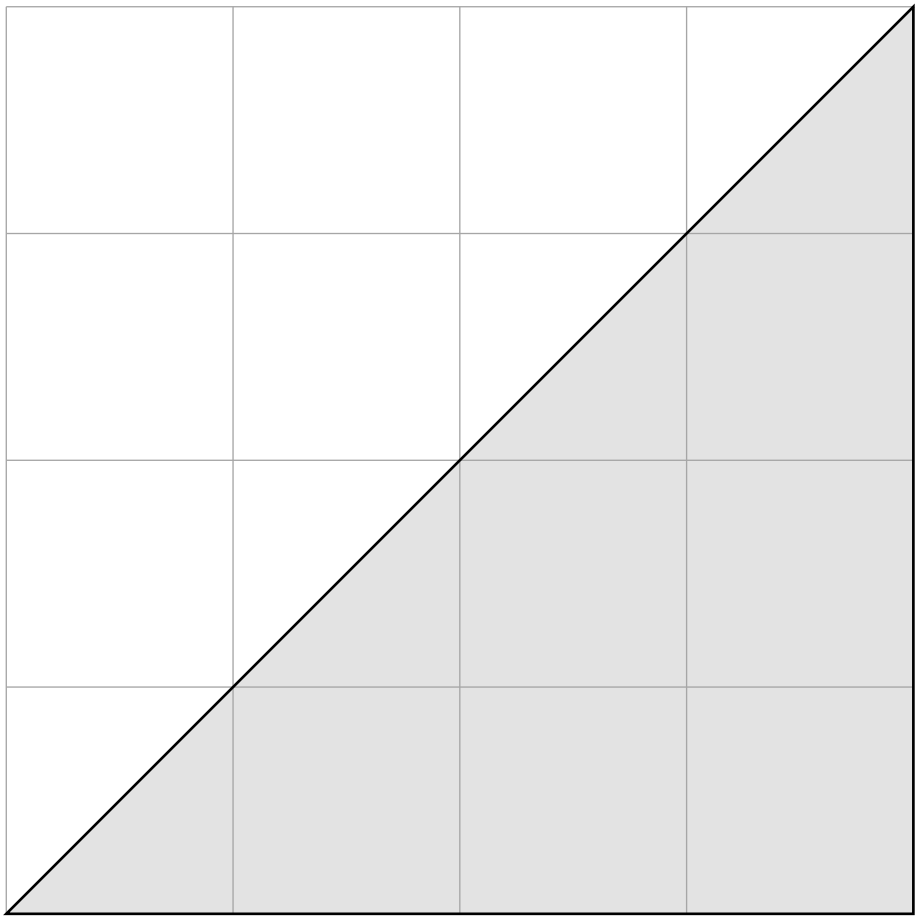
Whole squares are all gray. A cover uses every square with some gray in it. A touch at an edge or corner adds no square.



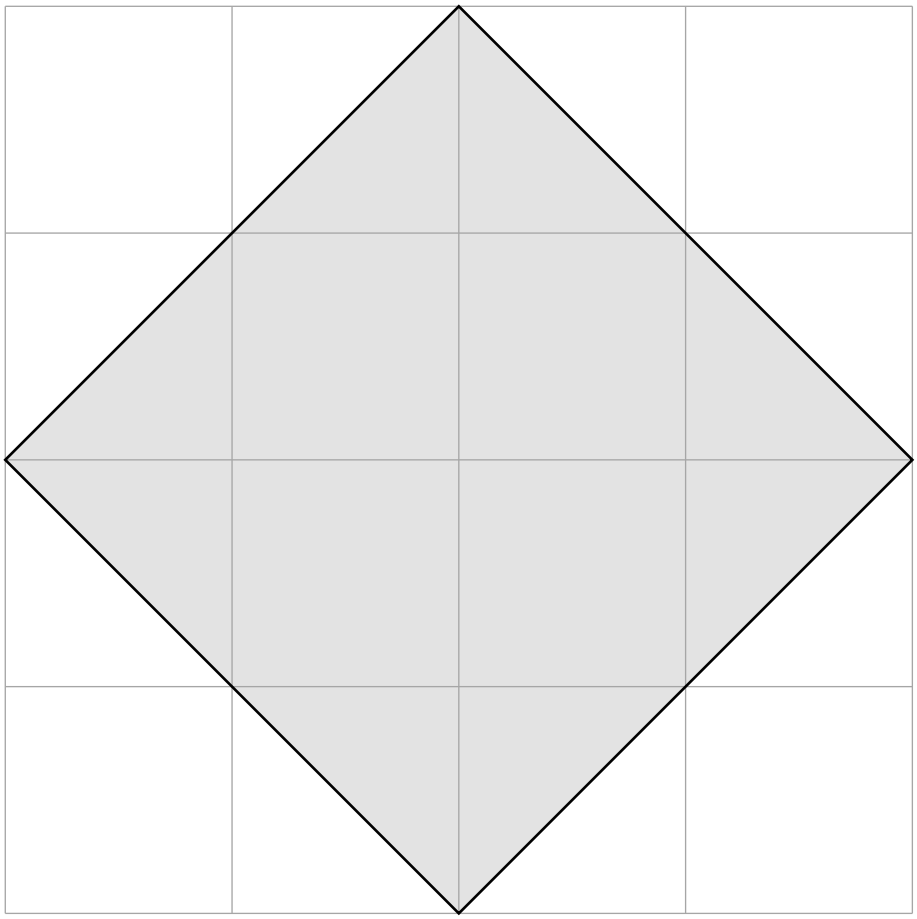
whole partial outside

Problem 1: Mark all the whole squares inside the triangle. Cover the triangle with as few grid-square tiles as you can.



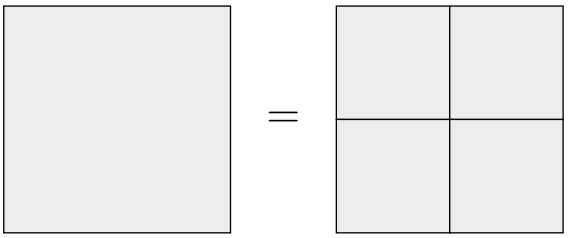
Inside: _____ big squares Cover: _____ big squares

Problem 2: Trap the diamond's area between its whole squares and its cover. Can you be certain its area is more than 3 big squares? Less than 13?



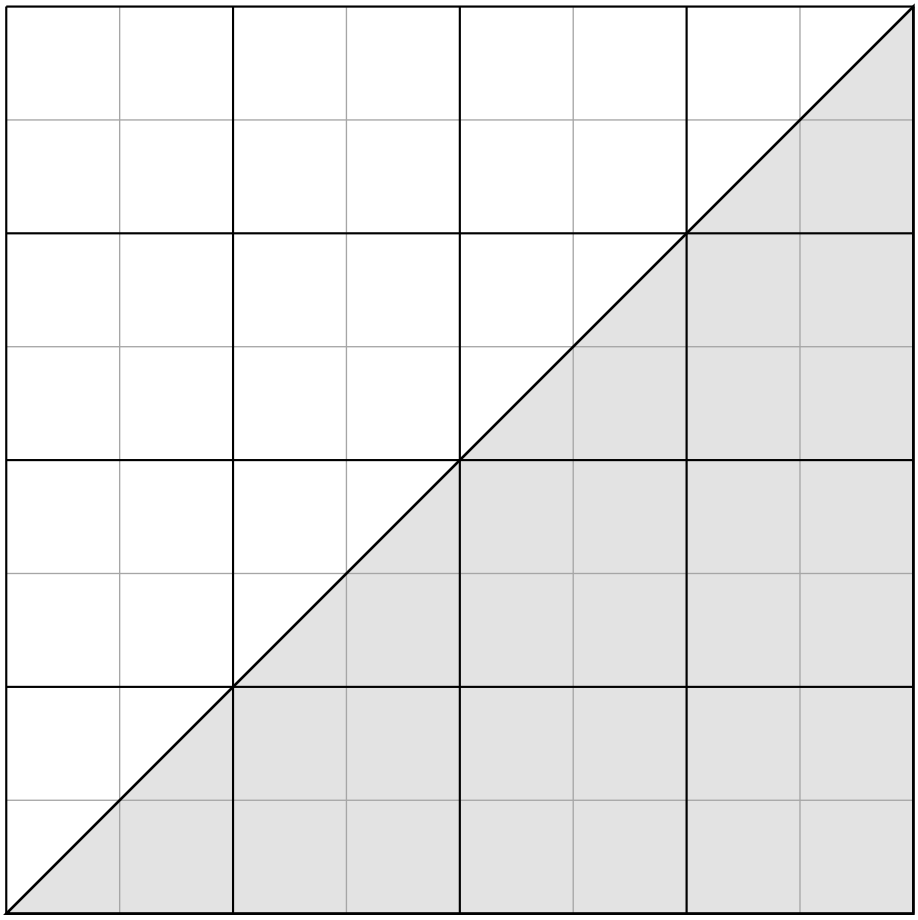
Inside: _____ big-square units

Cover: _____ big-square units



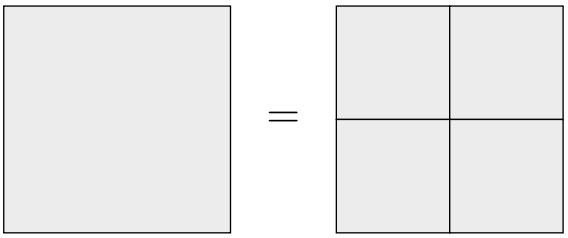
1 big square
= 4 small squares

Problem 3: Use the smaller squares to trap the triangle's area more tightly. State both bounds in big-square units.



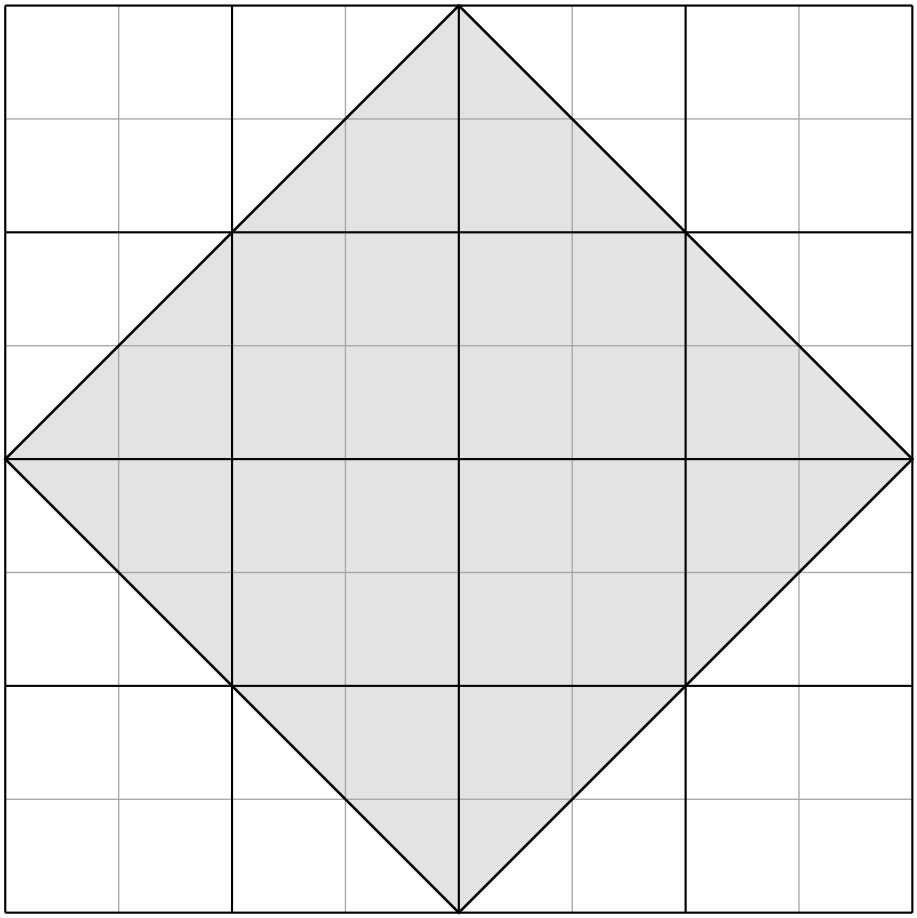
Inside: _____ big-square units

Cover: _____ big-square units



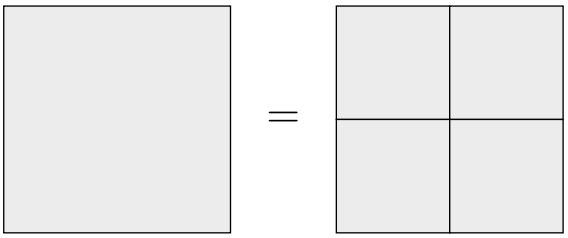
1 big square
= 4 small squares

Problem 4: Find new bounds for the diamond. Compare the area left uncertain on the big grid and on the small grid.



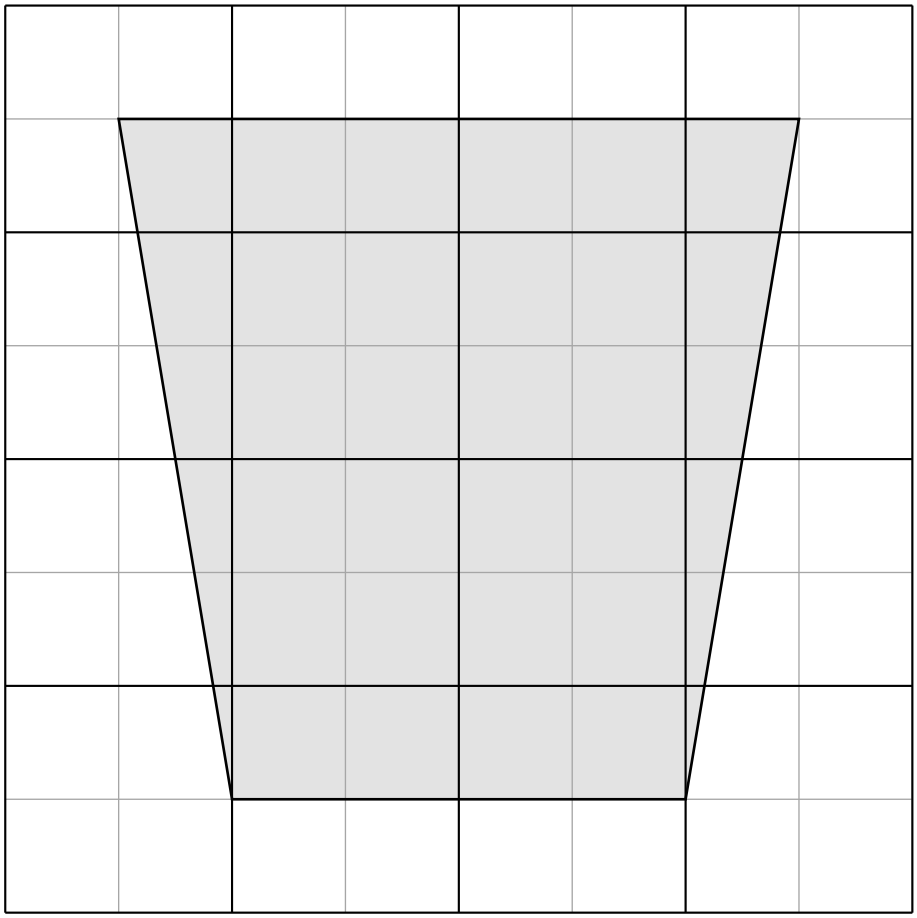
Inside: _____ big-square units

Cover: _____ big-square units



1 big square
= 4 small squares

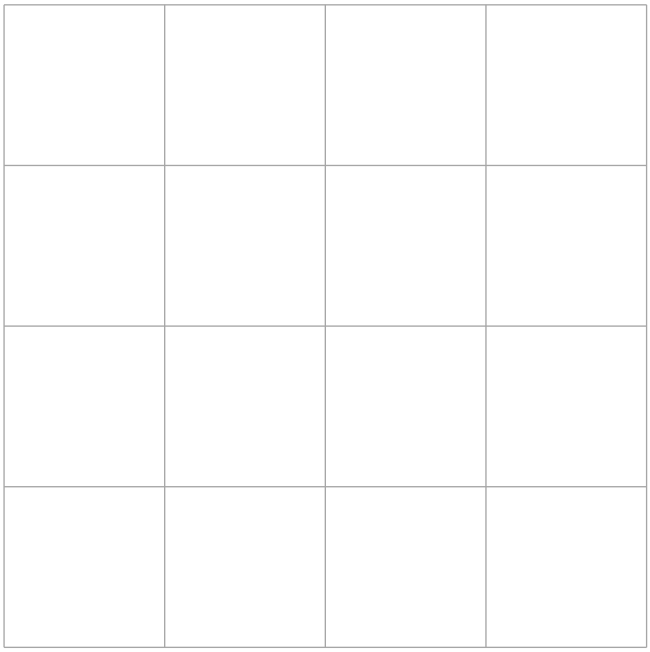
Problem 5: Trap this shape's area with big squares and then with small squares. Compare what happens to the area left uncertain with the triangle and the diamond.

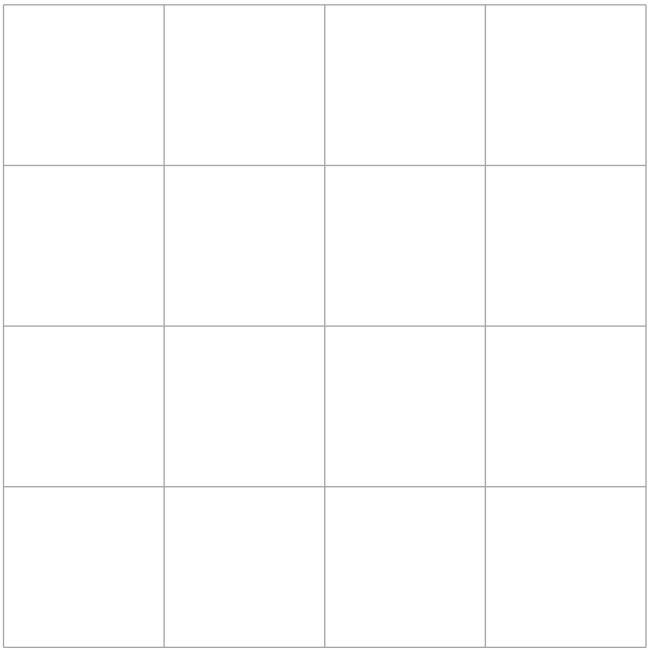


Big squares: Inside _____ Cover _____ big-square units

Small squares: Inside _____ Cover _____ big-square units

Problem 6: Draw two different shapes, each with exactly 4 whole gray squares and 4 partly gray squares. Make one area less than 5 big-square units and the other greater than 7.





Problem 7: Could any shape with exactly 4 whole and 4 partly gray squares have area less than 4 big squares? More than 8? Decide what the square markings guarantee.